

$$\textcircled{1} m(a+bX) = a + b \times m(X) \rightarrow \text{multiply to simplify}$$

$$m(a+bX) = \frac{1}{N} \sum_{i=1}^N (bx_i + a) \rightarrow m(X) = \frac{1}{N} \sum_{i=1}^N x_i$$

$$m(a+bX) = \frac{1}{N} \left(\sum_{i=1}^N bx_i + \sum_{i=1}^N a \right)$$

$$m(a+bX) = \frac{1}{N} \left(b \sum_{i=1}^N x_i + Na \right) \rightarrow \text{distribute to simplify}$$

$$m(a+bX) = \frac{1}{N} b \sum_{i=1}^N x_i + a \rightarrow \text{rearrange}$$

$$m(a+bX) = a + b \left(\frac{1}{N} \sum_{i=1}^N x_i \right) \rightarrow \text{Same as } m(X)$$

$$m(a+bX) = a + bm(X)$$

$\textcircled{4}$ NO, because the mean (average) changes, thus leading to a different result when it is recorded before vs after transformation

$$\textcircled{2} \text{cov}(X, a + bY) = b \times \text{cov}(X, Y)$$

$$\text{cov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (\cancel{a} + by_i - (\cancel{a} + bm(Y)))$$

$$\text{cov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (by_i - bm(Y)) \rightarrow \text{pull the 'b' out}$$

$$\text{cov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) b(y_i - m(Y)) \rightarrow \text{Same as } b \text{cov}(X, Y)$$

$$\textcircled{3} \text{cov}(a + bX, a + bX) = b^2 \text{cov}(X, X)$$

$$\text{cov}(a + bX, a + bX) = \frac{1}{N} \sum_{i=1}^N ((a + bx_i) - m(a + bX)) ((a + bx_i) - m(a + bX))$$

$$\text{cov}(a + bX, a + bX) = \frac{1}{N} \sum_{i=1}^N [(a + bx_i) - (a + bm(X))]^2$$

$$\text{cov}(a + bX, a + bX) = \frac{1}{N} \sum_{i=1}^N [b(x_i - m(X))]^2$$

$$\text{cov}(a + bX, a + bX) = \frac{1}{N} \sum_{i=1}^N b^2 (x_i - m(X))^2$$

$$\text{cov}(a + bX, a + bX) = b^2 \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2 \rightarrow \text{Same as } \text{cov}(X, X) = \dots$$

$$\text{cov}(a + bX, a + bX) = b^2 \text{cov}(X, X)$$

$$\ast \text{cov}(X, X) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

$$\text{cov}(X, X) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (x_i - m(X))$$

$$\text{cov}(X, X) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2 \rightarrow \text{Same as } S^2, \text{ thus } \text{cov}(X, X) = S^2$$